A Multi-Agent System for Preventing Environmentally Harmful Activities

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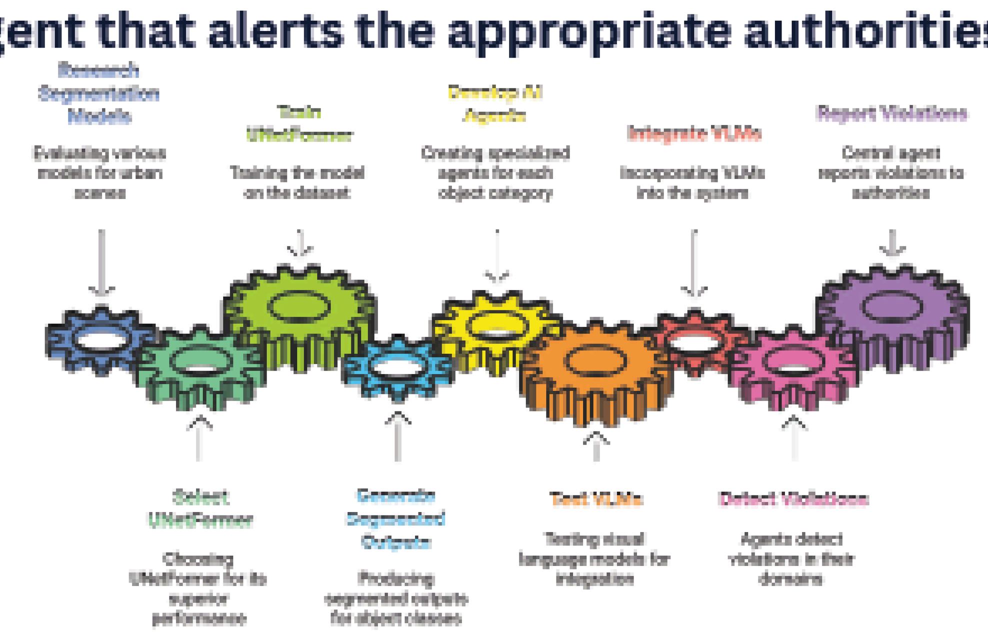
Introduction

- Urban violations—like illegal construction, dumping, and pollution—are growing and often go undetected.
- Current systems are manual, fragmented, and lack real-time coordination.
- There is no unified system that connects detection and reporting across multiple object types.
- This research proposes an AI-powered solution using segmentation, VLMs, and specialized agents to enable real-time, connected monitoring.

Methodology

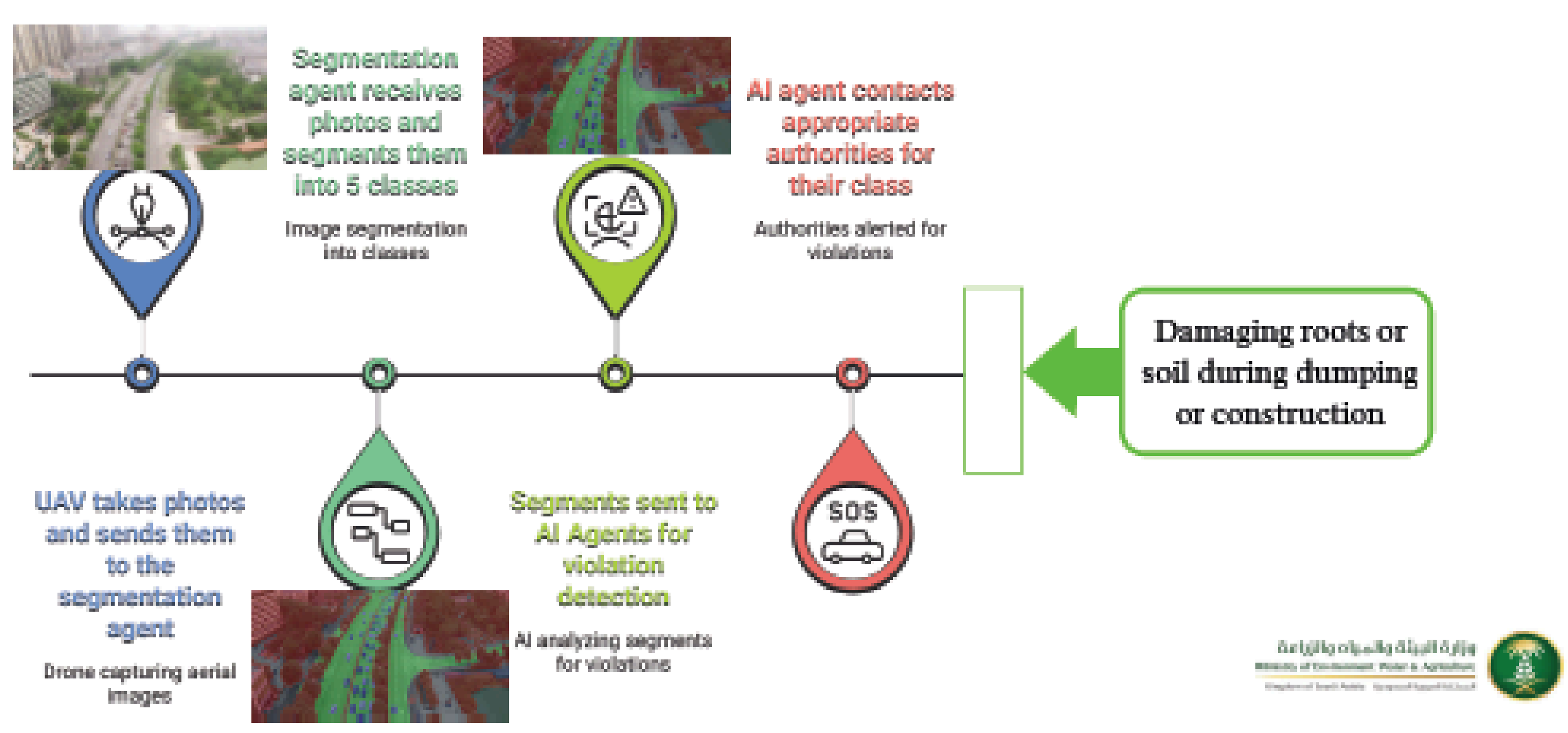
We began by evaluating top segmentation models for urban scenes, including ABCNet, UAVID, BiSeNet, MSD, UNetFormer, and SAM, using the UAVID dataset. Based on the study "A combined super-resolution and semantic segmentation approach of urban landscape image in rain and fog environment," UNetFormer showed the highest accuracy. Although we tested SAM, we ultimately selected UNetFormer for its superior performance in complex environments.

After training UNetFormer on our dataset, we generated accurate segmentation outputs. We are now developing object-specific AI agents and testing various visual language models (VLMs). Each agent will detect violations within its class and forward them to a central AI agent that alerts the appropriate authorities in real time.



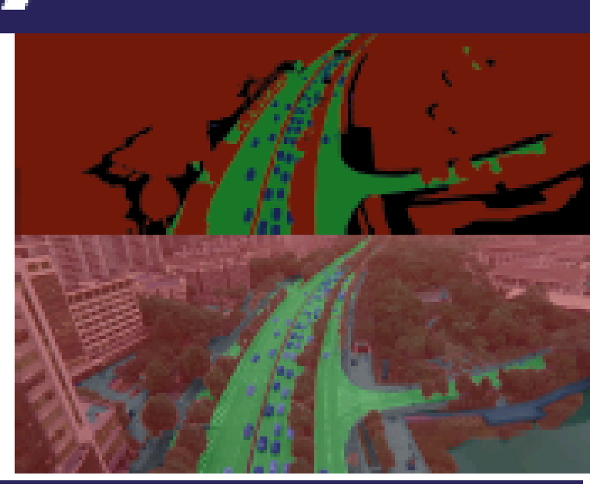
Results

	TRAIN	TEST
F1	92.79%	86.80%
Overall Accuracy	95.94%	92.34%
mIoU	86.83%	77.21%
epochs	39	
trainer/global_step	15999	



Data Analysis

The model performed well across all metrics, with an F1 Score of 86.80%, Overall Accuracy of 92.34%, and mIoU of 77.21% on the test set. These results indicate strong generalization, accurate segmentation, and effective detection of object boundaries. The model is reliable and ready for integration into the full violation detection system.



Applications

- Pollution Detection and Control
- Illegal Activities Monitoring
- Disaster Prevention and Response
- Environmental Policy Enforcement
- Urban Environmental Management

Future work

Integrating vision-language models for better interpretation of visual data and adding a notification agent to alert relevant authorities in real time. The system will also be deployed in real-world environments to evaluate its effectiveness.

Conclusion

This research set out to answer whether an AI-powered, connected system could accurately detect and report urban violations across multiple object types. By evaluating several segmentation models, we identified UNetFormer as the most effective for complex urban scenes. After training, the model achieved high accuracy and strong generalization, with an F1 Score of 86.80% and mIoU of 77.21% on test data.

Currently, we are integrating specialized AI agents and testing visual language models to enable real-time violation detection and reporting. The system's modular design and reliable performance demonstrate its potential to fill the existing gap in disconnected urban monitoring tools.

References

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